STRONGLY CARTESIAN LIFTS OF A COVERING DEFINE A TOPOLOGY ON A FIBRED CATEGORY

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Composition of strongly cartesian morphisms. categories-lemma-composition-cartesian (cross-chapter, named). Gives that isomorphisms and compositions of strongly cartesian morphisms are strongly cartesian, used in all three axioms.
- Representable fibres go up. categories-lemma-fibred-category-representable-goes-up (cross-chapter, named). Produces the base-changed family x_i x_x y with strongly cartesian projection and computes its image in C.

Source reference. Original reference 02XL: focused-seeds/geometry/S9-stk-topology-inherited/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06N5: focused-seeds/geometry/S9-stk-topology-inherited/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00VH: focused-seeds/geometry/S9-stk-topology-inherited/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $p : \mathcal{S} \to \mathcal{C}$ be a fibred category. Let $Cov(\mathcal{S})$ be the set of families $\{x_i \to x\}_{i \in I}$ of morphisms in $\mathcal{S}$ with fixed target such that (a) each $x_i \to x$ is strongly cartesian, and (b) $\{p(x_i) \to p(x)\}_{i \in I}$ is a covering of $\mathcal{C}$. Then $(\mathcal{S}, Cov(\mathcal{S}))$ is a site.*

Proof. We have to check the three conditions of Sites, Definition 00VH.

(1) If $x \to y$ is an isomorphism of $\mathcal{S}$, then it is strongly cartesian by Categories, Lemma 02XL and $p(x) \to p(y)$ is an isomorphism of $\mathcal{C}$. Thus $\{p(x) \to p(y)\}$ is a covering of $\mathcal{C}$ whence $\{x \to y\} \in \mathrm{Cov}(\mathcal{S})$.

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(2) If $\{x_i \to x\}_{i \in I} \in \mathrm{Cov}(\mathcal{S})$ and for each i we have $\{y_{ij} \to x_i\}_{j \in J_i} \in \mathrm{Cov}(\mathcal{S})$, then each composition $y_{ij} \to x$ is strongly cartesian by Categories, Lemma 02XL and $\{p(y_{ij}) \to p(x)\}_{i \in I, j \in J_i} \in \mathrm{Cov}(\mathcal{C})$. Hence also $\{y_{ij} \to x\}_{i \in I, j \in J_i} \in \mathrm{Cov}(\mathcal{S})$.
(3) Suppose $\{x_i \to x\}_{i \in I} \in \mathrm{Cov}(\mathcal{S})$ and $y \to x$ is a morphism of $\mathcal{S}$. As $\{p(x_i) \to p(x)\}$ is a covering of $\mathcal{C}$ we see that $p(x_i) \times_{p(x)} p(y)$ exists. Hence Categories, Lemma 06N5 implies that $x_i \times_x y$ exists, that $p(x_i \times_x y) = p(x_i) \times_{p(x)} p(y)$, and that $x_i \times_x y \to y$ is strongly cartesian. Since also $\{p(x_i) \times_{p(x)} p(y) \to p(y)\}_{i \in I} \in \mathrm{Cov}(\mathcal{C})$ we conclude that $\{x_i \times_x y \to y\}_{i \in I} \in \mathrm{Cov}(\mathcal{S})$

This finishes the proof. □